Medication Reminder System – Secure Serverless Healthcare App

Built on AWS for HIPAA-Compliant Cloud Healthcare

Problem Statement

Patients often forget medications → poor treatment adherence.

Need for an automated, secure, cloud-based reminder system.

Must protect patient privacy and meet HIPAA standards.

Project Objective

Build a serverless AWS web app to:

- Manage medication schedules and reminders.
- Notify users automatically via SNS.
- Secure all data with encryption and MFA.

AWS Services Used

Authentication	Amazon Cognito	Secure user sign-in + MFA
Compute	AWS Lambda	Run backend logic serverlessly
API Layer	Amazon API Gateway	REST API endpoints
Database	Amazon DynamoDB	Store meds & reminders
Storage	Amazon S3	Host static web UI
CDN / Edge	Amazon CloudFront	Global HTTPS delivery
Notifications	Amazon SNS,Amazon EventBridge	Send reminders & alerts
Security	AWS WAF, CloudTrail, CloudWatch	Threat protection & monitoring

Architecture Overview

User → WAF →CloudFront → S3 (Web UI) → API Gateway → Lambda → DynamoDB → SNS (Reminders)
Security Layer: WAF → [App] Logging & Monitoring: CloudTrail → CloudWatch → SNS Alerts

Edge Delivery

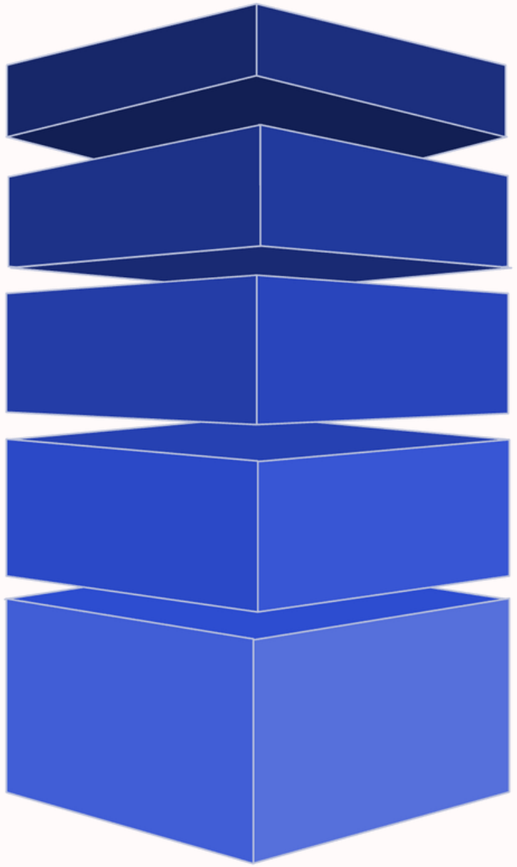
User to CloudFront for global caching.

Compute

API Gateway invokes Lambda functions.

Integrations

Side paths to S3 and SNS; shield and magnifier overlay.



API Layer

CloudFront → API Gateway to route requests.

Data Store

Lambda reads/writes to DynamoDB.

Amazon S3: Secure Storage & Web Hosting

Hosts the entire frontend web application (HTML, CSS, JS).

Configured for Static Website Hosting with CloudFront CDN → ensures fast, global access.

HTTPS-only bucket policy blocks insecure (HTTP) requests.

Server-Side Encryption (SSE-S3, AES-256) protects all stored assets and audit logs.

Stores CloudTrail and WAF logs for compliance evidence.

Amazon DynamoDB: Medication Data Storage

1

Primary database for all medication and reminder data.

Tables used:

- Medications → user_id (PK), medication_id (SK)
- Reminders → reminder_id (PK)
- ReminderLogs → user_id (PK), log_id (SK)

2

Lambda functions perform CRUD operations through the API Gateway.

3

AES-256 encryption at rest + fine-grained IAM access control.

4

Fully serverless → auto-scales with zero maintenance cost.

Security Architecture

WAF
Blocks SQL Injection, XSS, DDoS.

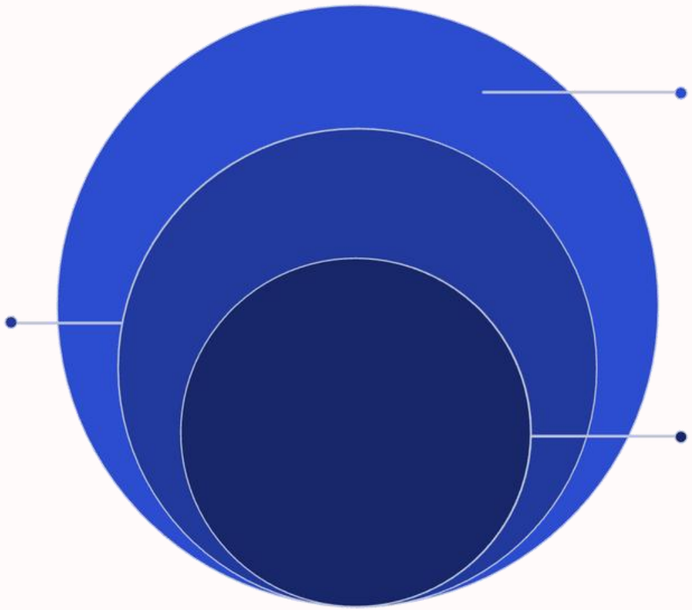
CloudFront
TLS 1.2 encryption + HTTPS redirect.

Cognito
MFA authentication.

API Gateway
Throttling & validation.

Lambda + DynamoDB
AES-256 encryption at rest.

CloudTrail + CloudWatch
Continuous auditing & alerts.



Edge & Protection
WAF shield, CloudFront, Cognito, API Gateway

Compute & Data
Lambda with DynamoDB AES-256 encrypted

Core Audit
CloudTrail + CloudWatch continuous monitoring

How Each AWS Service Enhances Security

Cognito	MFA, password policies, JWT tokens
Lambda	Encrypted environment variables
S3	HTTPS-only policy, SSE-S3 encryption
DynamoDB	AES-256 encryption at rest
CloudTrail	API activity auditing + tamper validation
CloudWatch	Metric filters + real-time alarms
WAF	Blocks OWASP Top 10 threats
SNS	Secure notification delivery (TLS)

Monitoring and Alerts

- CloudWatch Metric Filter: detects AccessDenied, UnauthorizedAccess.
- Alarm: Triggers SNS topic → email alert.
- CloudTrail: records all API calls with KMS encryption.

Ensures rapid detection & response to security incidents.

Cost Optimization

MFA (TOTP)	0.00
Encryption (all services)	0.00
CloudTrail Audit Logs	0.00
CloudWatch + SNS Alerts	0.50
AWS WAF (Web ACL + Rules)	7.00
Total Monthly Cost	8.00

HIPAA Compliance Mapping

HIPAA Requirement
Access Control

AWS Implementation
Cognito + MFA

HIPAA Requirement
Data Encryption

AWS Implementation
DynamoDB, S3, Lambda

HIPAA Requirement
Audit Controls

AWS Implementation
CloudTrail + Log Validation

HIPAA Requirement
Integrity

AWS Implementation
KMS Encryption Keys

HIPAA Requirement
Transmission Security

AWS Implementation
TLS 1.2 (HTTPS)

HIPAA Requirement
Monitoring & Alerts

AWS Implementation
CloudWatch + SNS

HIPAA Requirement
Threat Prevention

AWS Implementation
AWS WAF + Rate Limit

Conclusion & Future Scope

Key Takeaways

Secure, serverless, HIPAA-aligned healthcare application.

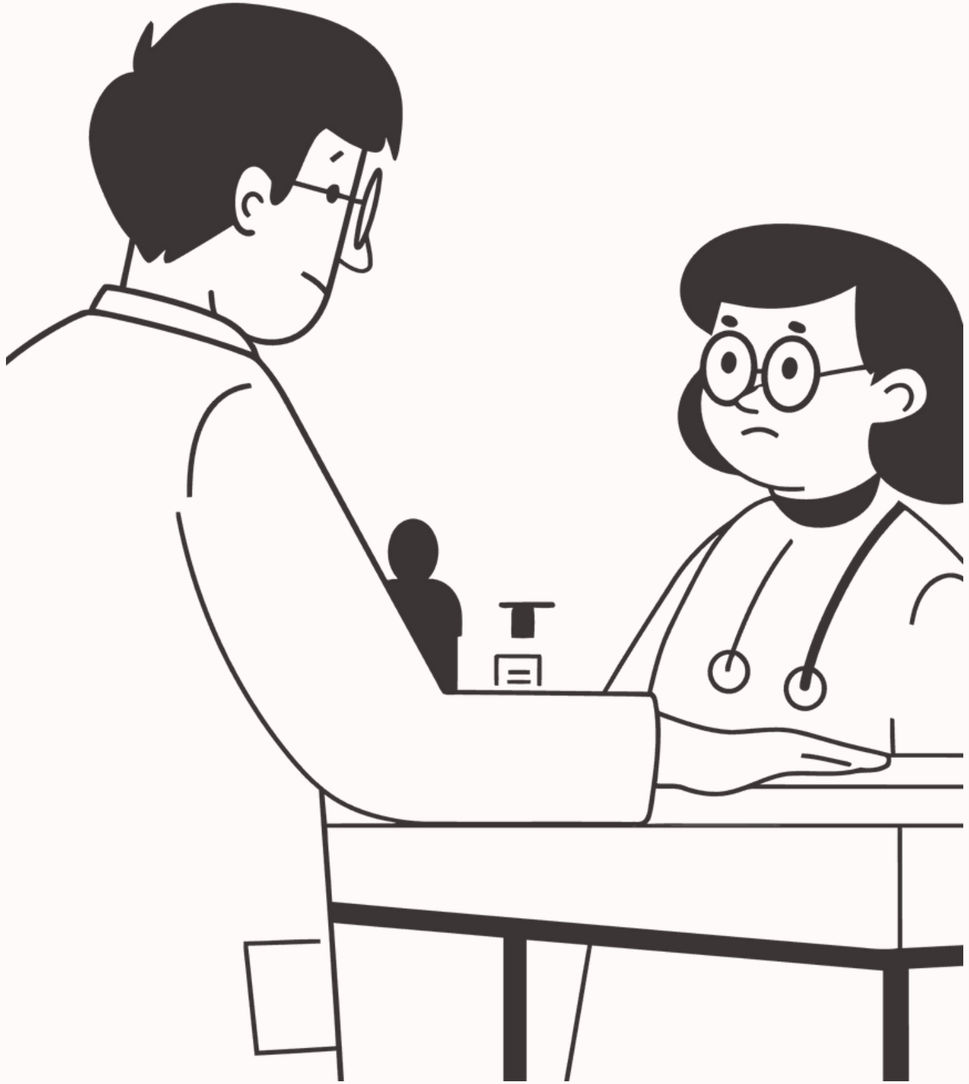
Demonstrates strong security architecture with low cost.

Future Enhancements:

AI-based medication prediction & analytics.

Notify users about potential health risks based on medication patterns

Dynamically adjust reminder timing based on user behavior



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Cloud Computing